

Aadiy Khan

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EDUCATION

Vellore Institute of Technology, Bhopal

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

CGPA: 8.55/10.0

EXPERIENCE

Summer Intern, Ordnance Clothing Factory (OCF), Shahjahanpur

May 2026 – July 2026

Ministry of Defence Summer Internship Program

Shahjahanpur, India

- Chosen among undergraduate applicants nationwide for a Ministry of Defence-sponsored internship focused on AI-driven textile defect detection.
- Building computer vision and deep learning pipelines using CNN-based classification to automatically detect tears, stains, stitching defects, and color inconsistencies on production-line fabric.
- Collaborating directly with defense production engineers and QA teams to translate model outputs into deployable, production-ready inspection workflows.

PROJECTS

Transformer From Scratch (Attention Is All You Need) [GitHub] | *PyTorch, NLP, LLMs, Deep Learning* 2026

- Implemented the complete Transformer architecture from the original “Attention Is All You Need” paper entirely from scratch in PyTorch, without relying on any high-level transformer libraries.
- Built Multi-Head Self-Attention, Positional Encoding, Layer Normalization, and Residual connections underpinning every modern LLM architecture, including masked attention for autoregressive decoding.
- Built end-to-end training and inference pipelines for sequence-to-sequence tasks, gaining first-principles understanding of the attention mechanisms and optimization techniques that power today’s LLMs.

FlowBERT: Multi-Task Customer Query Routing (NLP) [GitHub] | *TensorFlow, DistilBERT, NLP* Apr – May 2026

- Fine-tuned DistilBERT for simultaneous department classification and urgency prediction using a multi-task learning framework, cutting manual ticket triage effort by 75%.
- Designed a shared Transformer encoder backbone with task-specific classification heads, enabling efficient multi-output prediction from a single forward pass.
- Scaled the training corpus from 50 to 600+ labeled samples through targeted data augmentation and class-balancing, and deployed a real-time Streamlit app for inference and confidence visualization.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy

Deep Learning: CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms, Transfer Learning

NLP & LLMs: BERT, DistilBERT, HuggingFace Transformers, Embeddings, Text Classification, Sequence-to-Sequence Modeling

Tools & Deployment: Git, Docker, Streamlit, OpenCV, NLTK, Gensim

ACHIEVEMENTS

- Selected for OCF Shahjahanpur Summer Internship Program 2026 focused on AI-based defect detection for military clothing fabric.
- Reproduced the architecture proposed in the “Attention Is All You Need” paper, including Multi-Head Attention and Encoder-Decoder training pipelines.
- Solved 350+ algorithmic problems on LeetCode.
- IBM AI Engineer Professional Certificate.
- NPTEL Introduction to Machine Learning.

ADDITIONAL

Languages: Fluent in Hindi, English, and Urdu

Interests: Competitive Programming, Open Source Contributions, Badminton, Football